1. Evaluate the limit:

$$\lim_{x\to-\infty} \frac{x^2 - \sin(x)}{\sqrt{x^4+1}}$$

We know that:

$$-1 \leq \sin x \leq 1$$

$$1 \geq -\sin x \geq -1$$

$$-1 \leq -\sin x \leq 1$$

$$x^2 - 1 \leq x^2 - \sin x \leq x^2 + 1$$

$$\frac{x^2-1}{\sqrt{x^4+1}} \leq \frac{x^2-\sin x}{\sqrt{x^4+1}} \leq \frac{x^2+1}{\sqrt{x^4+1}}$$

$$\lim_{x\to-\infty} \frac{x^2-1}{\sqrt{x^4+1}} = \lim_{x\to-\infty} \frac{x^2\left(1-\frac{1}{x^2}\right)}{\sqrt{x^4\left(1+\frac{1}{x^4}\right)}} = \lim_{x\to-\infty} \frac{x^2\left(1-\frac{1}{x^2}\right)}{|x^2|\sqrt{1+\frac{1}{x^4}}}$$

$$|x^2| = x^2 \quad \forall x \in \mathbb{R}$$

$$= \lim_{x\to-\infty} \frac{x^2\left(1-\frac{1}{x^2}\right)}{x^2\sqrt{1+\frac{1}{x^4}}} = \lim_{x\to-\infty} \frac{1-\frac{1}{x^2}}{\sqrt{1+\frac{1}{x^4}}} = \frac{1-0}{\sqrt{1+0}}$$

$$= 1$$

$$\lim_{x\to-\infty} \frac{x^2+1}{\sqrt{x^4+1}} = \lim_{x\to-\infty} \frac{x^2\left(1+\frac{1}{x^2}\right)}{\sqrt{x^4\left(1+\frac{1}{x^4}\right)}} = \lim_{x\to-\infty} \frac{x^2\left(1+\frac{1}{x^2}\right)}{|x^2|\sqrt{1+\frac{1}{x^4}}}$$

$$|x^2| = x^2 \quad \forall x \in \mathbb{R}$$

$$= \lim_{x\to-\infty} \frac{x^2\left(1+\frac{1}{x^2}\right)}{x^2\sqrt{1+\frac{1}{x^4}}} = \lim_{x\to-\infty} \frac{1+\frac{1}{x^2}}{\sqrt{1+\frac{1}{x^4}}} = \frac{1+0}{\sqrt{1+0}}$$

$$= 1$$

By Squeeze Theorem:

$$\lim_{x \to -\infty} \frac{x^2 - \sin x}{\sqrt{x^4+1}} = 1$$

2. Determine the value of a and b so that the function f is continuous everywhere.

$$f(x) = \begin{cases} ax - 3, & x < 1, \\ \frac{bx-1}{2}, & 1 \le x \le 3, \\ \frac{x^2 - (b-2)x + a - 4}{x-2}, & x > 3. \end{cases}$$

Let $h(x) = \dfrac{x^2 - (b-2)x + a - 4}{x-2}$, $x > 3$ or $(3, +\infty)$

dom $h = \mathbb{R} - \{x = 2\}$ but $x = 2 \notin (3, +\infty)$ so $x = 2$ is not a candidate for possible points of discontinuity.

Candidates for points of discontinuity:

$x = 1$, $x = 3$

at $x = 1$:

$$f(1) = \frac{b(1) - 1}{2} = \frac{b-1}{2}$$

Taking limits from both sides:

$$\lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} \frac{bx-1}{2} = \frac{b(1)-1}{2} = \frac{b-1}{2}$$

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} (ax - 3) = a(1) - 3 = a - 3$$

$$\lim_{x \to 1^+} f(x) = \lim_{x \to 1^-} f(x)$$

$$\frac{b-1}{2} = a - 3 \implies b - 1 = 2a - 6$$

$$2a - b - 5 = 0 \quad (1)$$

at $x = 3$:

$$f(3) = \frac{b(3) - 1}{2}$$

$$f(3) = \frac{3b - 1}{2}$$

Taking limits from both sides:

$$\lim_{x \to 3^+} f(x) = \lim_{x \to 3^+} \frac{x^2 - (b-2)x + a - 4}{x - 2}$$

$$= \frac{(3)^2 - (b-2)(3) + a - 4}{3 - 2}$$

$$= \frac{9 - 3b + 6 + a - 4}{1}$$

$$= a - 3b + 11$$

$$\lim_{x \to 3^-} f(x) = \lim_{x \to 3^-} \frac{bx - 1}{2}$$

$$= \frac{3b - 1}{2}$$

$$\lim_{x \to 3^+} f(x) = \lim_{x \to 3^-} f(x)$$

$$a - 3b + 11 = \frac{3b - 1}{2}$$

$$2a - 6b + 22 = 3b - 1$$

$$2a - 9b + 23 = 0 \quad (2)$$

$$\begin{cases} 2a - b - 5 = 0 & (1) \\ 2a - 9b + 23 = 0 & (2) \end{cases}$$

$(1) - (2)$:

$$(2a - b - 5) - (2a - 9b + 23) = 0$$

$$8b - 28 = 0$$

$$b = \frac{7}{2}$$

$$2a - b - 5 = 0$$

$$2a - \frac{7}{2} - 5 = 0$$

$$2a = \frac{7}{2} + 5$$

$$2a = \frac{17}{2}$$

$$a = \frac{17}{4}$$

$$\therefore \boxed{a = \frac{17}{4} \quad b = \frac{7}{2}}$$

3. Evaluate the limit:

$$\lim_{x\to 5^-} \frac{|x-5|}{2x^2-11x+5}$$

Definition of absolute value:

$$|x-5| = \begin{cases} x-5, & x \geq 5 \\ -(x-5), & x < 5 \end{cases}$$

$x \to 5^- \Rightarrow x < 5$

Thus,

$$\lim_{x\to 5^-} \frac{|x-5|}{2x^2-11x+5} = \lim_{x\to 5^-} \frac{-(x-5)}{2x^2-11x+5}$$

$$= \lim_{x\to 5^-} \frac{-(x-5)}{(x-5)(2x-1)}$$

$$= \lim_{x\to 5^-} \frac{-1}{2x-1}$$

$$= -\frac{1}{2(5)-1} = \boxed{-\frac{1}{9}}$$

4. Prove that

$$\lim_{x\to -2} (4x^2+7x+1) = 3$$

Proof:

$\lim_{x\to -2} (4x^2+7x+1) = 3$ iff $\forall \varepsilon > 0$, $\exists \delta > 0$ such that

$0 < |x+2| < \delta \Rightarrow |(4x^2+7x+1)-3| < \varepsilon$.

Observe that:

$$|(4x^2+7x+1)-3| = |4x^2+7x-2|$$

$$= |(x+2)(4x-1)|$$

$$= |x+2| \cdot |4x-1| < \varepsilon$$

Suppose $|x+2|<1$:

$-1<x+2<1$

$-3<x<-1$

$-12<4x<-4$

$-13<4x-1<-5$

$5<-(4x-1)<13$

$|-(4x-1)|<13$

We know that

$|x| = |-x|$

$|-(4x-1)| = |4x-1| < 13$

$\Rightarrow \quad 0<|x+2|<\delta$

$0<|4x-1|\cdot|x+2|<13\delta \quad (|4x-1|<13)$

Thus $13\delta = \varepsilon$

$\delta = \frac{\varepsilon}{13}$

Take $\delta = \min\left\{1, \frac{\varepsilon}{13}\right\}$

Check if $0<|x+2|<\delta \Rightarrow |(4x^2+7x+1)-3|<\varepsilon$.

$0<|x+2|<\delta$

$0<|4x+1|\cdot|x+2|<13\delta$

$0<|(4x^2+7x+1)-3|<13\left(\frac{\varepsilon}{13}\right)=\varepsilon$

$\therefore \lim_{x\to -2}(4x^2+7x+1)=3$ (Q.E.D.)

5. Determine the horizontal asymptote of the function:

$$f(x)=\frac{x^4-2\cos(2x)}{2\sqrt{x^8-2}+1}$$

The horizontal asymptotes of any function is:

$y = \lim_{x\to\infty} f(x)$ and $y = \lim_{x\to -\infty} f(x)$

as $x \to \infty$:

$$y = \lim_{x \to \infty} f(x)$$

$$y = \lim_{x \to \infty} \frac{x^4 - 2\cos 2x}{2\sqrt{x^8 - 2} + 1}$$

$$-1 \leq \cos 2x \leq 1$$

$$-2 \leq 2\cos 2x \leq 2$$

$$-2 \leq -2\cos 2x \leq 2$$

$$x^4 - 2 \leq x^4 - 2\cos 2x \leq x^4 + 2$$

$$\frac{x^4 - 2}{2\sqrt{x^8 - 2} + 1} \leq \frac{x^4 - 2\cos 2x}{2\sqrt{x^8 - 2} + 1} \leq \frac{x^4 + 2}{2\sqrt{x^8 - 2} + 1}$$

$$\lim_{x \to \infty} \frac{x^4 - 2}{2\sqrt{x^8 - 2} + 1} = \frac{1}{2} \quad \text{and} \quad \lim_{x \to \infty} \frac{x^4 - 2}{2\sqrt{x^8 - 2} + 1} = \frac{1}{2}$$

Use limit theorems!

By squeeze theorem:

$$\lim_{x \to \infty} \frac{x^4 - 2\cos 2x}{2\sqrt{x^8 - 2} + 1} = \frac{1}{2}$$

Thus, f has a horizontal asymptote at $y = \frac{1}{2}$

Similarly, as $x \to -\infty$,

$$\lim_{x \to -\infty} \frac{x^4 - 2\cos 2x}{2\sqrt{x^8 - 2} + 1} = \frac{1}{2}$$

(Repeating the previous process produces the same result)

Thus, f has a horizontal asymptote at $y = \frac{1}{2}$

$\therefore$ $y = \frac{1}{2}$ is the only horizontal asymptote of f.

6. Evaluate the limit:

$$\lim_{x \to 2} \frac{x^3 + x^2 - 4x - 4}{x - 2}$$

$$\lim_{x \to 2} \frac{x^3+x^2-4x-4}{x-2}$$

Factoring by grouping:

$$x^3+x^2-4x-4 = x^2(x+1) - 4(x+1)$$
$$= (x+1)(x^2-4)$$
$$= (x+1)(x+2)(x-2)$$

$$\lim_{x \to 2} \frac{x^3+x^2-4x-4}{x-2} = \lim_{x \to 2} \frac{(x+1)(x+2)(x-2)}{x-2}$$
$$= \lim_{x \to 2} (x+1)(x+2)$$
$$= (2+1)(2+2)$$
$$= 12$$

7. Let $f(x) = x^3 - x^2 + 2x - 5$. Use the Intermediate Value Theorem to show that there exists $c \in [0, 2]$ such that $f(c) = -3$ and find the value of c.

IVT: If f is continuous on $[a, b]$, $\exists\, c \in [a, b]$ such that $f(a) < f(c) < f(b)$.

Since f is a polynomial function, it is continuous $\forall x \in \mathbb{R}$. Thus, f is continuous on $[0, 2]$.

$f(0) = 0^3 - 0^2 + 2(0) - 5$ $\qquad$ $f(2) = 2^3 - 2^2 + 2(2) - 5$

$f(0) = -5$ $\qquad$ $f(2) = 3$

$f(0) < f(c) < f(2)$ ✓

$f(c) = -3$

$c^3 - c^2 + 2c - 5 = -3$

$c^3 - c^2 + 2c - 2 = 0$

$c^2(c-1) + 2(c-1) = 0$

$(c^2+2)(c-1) = 0$

$c^2 + 2 = 0$ (No solution)

$c - 1 = 0 \Rightarrow c = 1 \in [0, 2]$

$\boxed{c = 1}$

8. Suppose $f(x) = x^2 - 2x + 4$. Show that f takes an extreme value on $[0, 3]$ and determine its extreme values.

EVT: Suppose f is continuous on $[a, b]$, then f has an extreme value on $[a, b]$

Since f is a quadratic function, it is continuous everywhere Therefore it must be continuous on $[0, 3]$. By EVT, f has an extrema on $[0, 3]$

Since f is a quadratic function and a polynomial function, its possible extrema is at $x = h$ where h is the x-coordinate of its vertex.

$f(x) = x^2 - 2x + 4$

$f(x) = x^2 - 2x + 1 + 3$

$f(x) = (x-1)^2 + 3$

Vertex: $(1, 3)$

$(1, 3)$ is the absolute minimum since the leading coefficient of f is positive and $1 \in [0, 3]$

Since the end behavior of f is $+\infty$ as $x \to \pm\infty$, it has no absolute maxima.

$f(0) = 0^2 - 2(0) + 4$ $\quad$ $f(3) = 3^2 - 2(3) + 4$

$f(0) = 4$ $\quad$ $f(3) = 7$

Since $f(3) > f(0)$, f has a local maxima at $(3,7)$

Absolute minima: $(1,3)$
Local maxima: $(3,7)$

9. Evaluate the limit:

$$\lim_{x\to 0} \frac{e^{4x} - e^{3x}}{5x}$$

$$\lim_{x\to 0} \frac{e^{4x} - e^{3x}}{5x} = \lim_{x\to 0} \frac{e^{3x}(e^x - 1)}{5x}$$

$$= \lim_{x\to 0} \frac{e^{3x}(e^x - 1)}{5x} \cdot \frac{\frac{1}{5}}{\frac{1}{5}}$$

$$= \lim_{x\to 0} \frac{e^{3x}}{5} \cdot \frac{e^x - 1}{x}$$

$$= \lim_{x\to 0} \frac{e^{3x}}{5} \cdot \lim_{x\to 0} \frac{e^x - 1}{x}$$

$$= \left(\frac{e^{3(0)}}{5}\right)(1) = \boxed{\frac{1}{5}}$$

10. Evaluate the limit:

$$\lim_{x\to 5^+} \left(\frac{1}{x-5}\right)\left(\frac{x-1}{4} - 1\right)$$

$$\lim_{x\to 5^+} \left(\frac{1}{x-5}\right)\left(\frac{x-1}{4} - 1\right) = \infty \cdot 0$$

$$= \lim_{x\to 5^+} \left(\frac{1}{x-5}\right)\left(\frac{x-1}{4} - \frac{4}{4}\right)$$

$$= \lim_{x\to 5^+} \left(\frac{1}{x-5}\right)\left(\frac{x-5}{4}\right)$$

$$= \lim_{x\to 5^+} \left(\frac{1}{4}\right) = \boxed{\frac{1}{4}} \qquad \frac{x-5}{x-5} = 1$$

11. Determine the horizontal asymptote of the function:
$f(x) = \dfrac{2[\![x]\!]+1}{x^2}$. Where $[\![x]\!]$ denotes the greatest integer x

as $x \to \infty$:

$$x \le [\![x]\!] < x+1$$
$$2x \le 2[\![x]\!] < 2x+2$$
$$2x+1 \le 2[\![x]\!]+1 < 2x+3$$
$$\frac{2x+1}{x^2} \le \frac{2[\![x]\!]+1}{x^2} < \frac{2x+3}{x^2}$$

$\lim_{x\to\infty} \frac{2x+1}{x^2} = 0$

$\lim_{x\to\infty} \frac{2x+3}{x^2} = 0$

} Use limit theorems

By Squeeze Theorem:

$$\lim_{x\to\infty} \frac{2[\![x]\!]+1}{x^2} = 0$$

as $x \to -\infty$:

$$\lim_{x\to-\infty} \frac{2[\![x]\!]+1}{x^2} = 0$$

(Repeating the previous process produces the same result)

$\therefore$ $y=0$ is the horizontal asymptote of f.

12. Evaluate the limit:

$$\lim_{x\to 2} \frac{\sin(x^2-4)}{5x-10}$$

$$\lim_{x\to 2} \frac{\sin(x^2-4)}{5x-10} = \lim_{x\to 2} \frac{\sin[(x+2)(x-2)]}{5(x-2)}$$

$$= \lim_{x\to 2} \frac{\sin[(x+2)(x-2)]}{5(x-2)} \cdot \frac{x+2}{x+2}$$

$$= \lim_{x\to 2} \frac{(x+2)\sin(x^2-4)}{5(x^2-4)}$$

$$= \lim_{x\to 2} \frac{x+2}{5} \cdot \lim_{x\to 2} \frac{\sin(x^2-4)}{x^2-4}$$

$$= \left(\frac{2+2}{5}\right)(1) = \boxed{\frac{4}{5}}$$

13. Determine the value(s) of x that make the function f discontinuous and identify the type of discontinuity:

$$f(x) = \begin{cases} 3x+4, & x \le -1, \\ \dfrac{x-3}{x^2-x-6}, & -1 < x \le 4 \\ \dfrac{x-5}{x^2-3x-10}, & x > 4. \end{cases}$$

Let $P(x) = 3x+4, \quad x \le -1$

$$g(x) = \frac{x-3}{x^2-x-6}, \quad -1 < x \le 4 \Rightarrow g(x) = \frac{x-3}{(x-3)(x+2)}, \quad -1 < x \le 4$$

$$h(x) = \frac{x-5}{x^2-3x-10}, \quad x > 4 \Rightarrow h(x) = \frac{x-5}{(x-5)(x+2)}, \quad x > 4$$

Possible points of discontinuity

$x = \{-1, 3, 4, 5\}$

at $x = -1$:

$f(-1) = 3(-1) + 4$

Using one sided limits:

$$\lim_{x \to -1^+} f(x) = \lim_{x \to -1^+} \frac{x-3}{x^2 - x - 6}$$

$$= \lim_{x \to -1^+} \frac{x-3}{(x-3)(x+2)}$$

$$= \lim_{x \to -1^+} \frac{1}{x+2}$$

$$= \frac{1}{-1+2}$$

$$= 1$$

$$\lim_{x \to -1^-} f(x) = \lim_{x \to -1^-} (3x+4)$$

$$= 3(-1) + 4$$

$$= 1$$

$$\lim_{x \to -1} f(x) = 1$$

Since $f(-1) = \lim_{x \to -1} f(x) = 1$, f is continuous at $x = -1$.

at $x = 3$:

$$f(3) = \frac{3-3}{3^2 - 3 - 6} = \frac{0}{0}$$

One sided limits:

$$\lim_{x \to 3^+} f(x) = \lim_{x \to 3^+} \frac{x-3}{x^2 - x - 6}$$

$$= \lim_{x \to 3^+} \frac{x-3}{(x-3)(x+2)}$$

$$= \lim_{x \to 3^+} \frac{1}{x+2} = \frac{1}{3+2} = \frac{1}{5}$$

$$\lim_{x\to 3^-} f(x) = \lim_{x\to 3^-} \frac{x-3}{x^2-x-6}$$

$$= \lim_{x\to 3^-} \frac{x-3}{(x-3)(x+2)}$$

$$= \lim_{x\to 3^-} \frac{1}{x+2} = \frac{1}{3+2} = \frac{1}{5}$$

$$\therefore \lim_{x\to 3} f(x) = \frac{1}{5}$$

Since $\lim_{x\to 3} f(x)$ exists and $f(3)$ is not defined, f has a removable discontinuity at $x=3$.

at $x=4$:

$$f(4) = \frac{4-3}{4^2-4-6}$$

$$f(4) = \frac{1}{6}$$

Using one sided limits:

$$\lim_{x\to 4^+} f(x) = \lim_{x\to 4^+} \frac{x-5}{x^2-3x-10}$$

$$= \lim_{x\to 4^+} \frac{x-5}{(x-5)(x+2)}$$

$$= \lim_{x\to 4^+} \frac{1}{x+2}$$

$$= \frac{1}{4+2}$$

$$= \frac{1}{6}$$

$$\lim_{x\to 4^-} f(x) = \lim_{x\to 4^-} \frac{x-3}{x^2-x-6}$$

$$= \lim_{x\to 4^-} \frac{x-3}{(x-3)(x+2)}$$

$$= \lim_{x\to 4^-} \frac{1}{x+2}$$

$$= \frac{1}{4+2}$$

$$= \frac{1}{6}$$

$\therefore \lim_{x\to 4} f(x) = \frac{1}{6}$

Since $f(4) = \lim_{x\to 4} f(x) = \frac{1}{6}$, f is continuous at $x = 4$

at $x = 5$:

$$f(5) = \frac{5-5}{5^2-3(5)-10} = \frac{0}{0}$$

Using one-sided limits:

$$\lim_{x\to 5^+} f(x) = \lim_{x\to 5^+} \frac{x-5}{x^2-3x-10}$$

$$= \lim_{x\to 5^+} \frac{x-5}{(x-5)(x+2)}$$

$$= \lim_{x\to 5^+} \frac{1}{x+2}$$

$$= \frac{1}{5+2} = \frac{1}{7}$$

$$\lim_{x\to 5^-} f(x) = \lim_{x\to 5^-} \frac{x-5}{x^2-3x-10}$$

$$= \lim_{x\to 5^-} \frac{x-5}{(x-5)(x+2)}$$

$$= \lim_{x \to 5^-} \frac{1}{x+2}$$

$$= \frac{1}{5+2} = \frac{1}{7}$$

$\therefore \lim_{x \to 5} f(x) = \frac{1}{7}$

Since $\lim_{x \to 5} f(x)$ exists and $f(5)$ is undefined, f has a removable discontinuity at $x = 5$.

Points of discontinuities:

$x = 3$, removable

$x = 5$, removable

14. Evaluate the limit:

$$\lim_{x\to 6}\frac{\sqrt[3]{x-5}-1}{x-6}$$

$$\lim_{x\to 6}\frac{\sqrt[3]{x-5}-1}{x-6}=\lim_{x\to 6}\frac{\sqrt[3]{x-5}-1}{x-6}\cdot\frac{\left(\sqrt[3]{x-5}\right)^2+\sqrt[3]{x-5}+1}{\left(\sqrt[3]{x-5}\right)^2+\sqrt[3]{x-5}+1}$$

$$=\lim_{x\to 6}\frac{\left(\sqrt[3]{x-5}\right)^3-1^3}{(x-6)\left(\left(\sqrt[3]{x-5}\right)^2+\sqrt[3]{x-5}+1\right)}$$

$$=\lim_{x\to 6}\frac{x-6}{(x-6)\left(\left(\sqrt[3]{x-5}\right)^2+\sqrt[3]{x-5}+1\right)}$$

$$=\lim_{x\to 6}\frac{1}{\left(\sqrt[3]{x-5}\right)^2+\sqrt[3]{x-5}+1}$$

$$=\frac{1}{\left(\sqrt[3]{6-5}\right)^2+\sqrt[3]{6-5}+1}$$

$$=\boxed{\frac{1}{3}}$$